Constructing a Medical Domain Functional Knowledge Graph with Large Language Models

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2024 IEEE 6th International Conference on Power, Intelligent Computing and Systems (ICPICS) | 979-8-3503-7431-5/24/$31.00 ©2024 IEEE | DOI: 10.1109/ICPICS62053.2024.10796042

***Abstract*—The integration of medicine and artificial intelligence is creating novel opportunities to gather, analyze, and generate groundbreaking medical insights from the vast expanse of medical literature. Although these advancements hold great promise, difficulties including manual annotation and precise data retrieval, and transparency persist. Advanced natural language processing (NLP) techniques and large language models (LLMs) have shown potential in addressing these issues. This paper introduces the Medical Knowledge Graph (MedKG), a comprehensive and multidisciplinary medical knowledge graph. By utilizing advanced NLP algorithms, MedKG extracts millions of entities and converts them into triples from a corpus of superior medical research papers published over the last decade. It categorizes unstructured medical information into nine distinct tags, including disease names, treatments, diagnoses, symptoms, genetics, cellular structures of diseases, patient data, medical procedures, and treatment applications, while seamlessly integrating the digital object identifier (DOI) of each paper. As the latest structured database of medical knowledge, MedKG serves as a powerful tool for accelerating medical research and lays the groundwork for constructing a more extensive medical knowledge graph using full-text medical articles. Moreover, our research establishes a foundation for practical knowledge management systems based on text mining, applicable not only to complex medical systems but also to other professional domains.**

Keywords—Medical engineering, NLP, Large language model, Knowledge graph, Database

I. INTRODUCTION

Modern medical research has made significant strides, yet the evolution of medical treatments often depends on traditional, labor-intensive trial-and-error processes combined with clinical expertise. These outdated methods slow down the advancement of cutting-edge medical therapies. The vast and specialized nature of medical science means that researchers in one field may find it challenging to access and interpret knowledge from other areas. For instance, those studying cancer treatments might not fully understand findings related to infectious diseases or neurological conditions[1]. Nevertheless, the biological mechanisms underlying various diseases are interrelated, and insights from one field can inform another. There is a pressing need to integrate insights from multiple medical disciplines to speed up research progress. However, essential medical knowledge is scattered across millions of scientific publications, encompassing topics like disease causes, diagnostic methods, treatment strategies, and the study of physiological, biochemical, and genetic factors, as well as their applications in pharmaceuticals, medical devices, and public health. This dispersion of knowledge creates obstacles to cross-disciplinary collaboration and innovation[2]. A major shortfall in the current medical research framework is the lack of a comprehensive database that can bring together this dispersed information, making it more accessible and promoting interdisciplinary work.

Even though databases like PubMed, Web of Science, and the Cochrane Library provide means to search for medical research papers using specific labels, extracting valuable medical information from the extensive body of literature continues to be a formidable challenge. To gain a clearer understanding of medical conditions and treatments, structured databases such as ClinVar, OMIM, and DrugBank have been developed[3]. However, these databases primarily contain data from clinical trials and genetic studies, which may not always fully align with real-world patient outcomes. Thus, the medical field urgently requires a database that leverages real patient data and clinical practice[4]. Furthermore, medical information in medicine is not limited to disease diagnosis and treatment options, but also includes patient demographics, comorbidities, and treatment outcomes. For example, patients with chronic conditions often require a personalized treatment plan based on their specific needs, medical history[5], and genetic characteristics. In this context, specific databases focused on patient outcomes and treatment effects have the potential to provide more valuable information to medical professionals and researchers than general medical databases such as PubMed that may not

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capture the nuances of personalized medicine[6]. This highlights the necessity of creating specialized databases to address the specific requirements of medical research and clinical practice, promoting more targeted approaches to disease treatment and research.

Knowledge graphs (KGs) are pivotal in the medical domain, offering structured representations of medical information[7], including controlled vocabulary and ontological relations. These graphs, comprised of nodes and edges, enable complex queries and insights crucial for medical research and decision-making. Despite their potential, constructing KGs in the medical field demands extensive input from domain experts due to the intricacies and specialized nature of medical knowledge[8]. This labor-intensive process hampers scalability, impacts performance, and poses challenges in maintaining timely and accurate data. The swift progress in natural language processing (NLP) has transformed the extraction of information from unstructured medical text, enabling the efficient and precise creation of medical knowledge graphs[9]. For example, the development of Medical Conditions Knowledge Graph (MCKG) has showcased the capability to integrate medical information from diverse sources, offering a unified view of medical knowledge, including disease etiology, diagnostic methods, treatment options, and genetic aspects. Specialized medical knowledge graphs such as Cardiovascular Disease Knowledge Graph (CVD-KG) and Neurological Disorders Knowledge Graph (ND-KG) have also emerged, focusing on specific medical areas and aiding in potential treatment discoveries.

However, constructing and maintaining medical knowledge graphs face significant challenges. Despite reduced dependency on experts with advancements in NLP, extensive annotation of training data remains essential to ensure accuracy and reliability[10]. Furthermore, predicting relationships between medical entities to form triples in the KG can sometimes lack authenticity, affecting the credibility of the graph. Updating medical knowledge graphs is another hurdle, as new medical entities require accurate relationship predictions with existing entities, complicating the graph's dynamic and accurate maintenance. To tackle these issues, the appearance of large language models (LLMs) like MedicalBERT and ClinicalBERT presents promising solutions. These LLMs, fine-tuned for medical text task, significantly enhance the extraction and credibility of structured medical information. We adopt such LLMs to improve tasks like Named Entity Recognition (NER) and Relation Extraction (RE) without extensive manual efforts, thus advancing medical knowledge graph construction and utilization in medical research and decision support systems.

In this research, we have made significant strides in developing a Medical Knowledge Graph (MedKG), an advanced graph database tailored specifically for medical research and knowledge organization. Our contributions are prominently featured in three main domains: (1) We present a novel approach for attaining exceptional precision in named entity recognition (NER), relation extraction (RE), and entity resolution (ER) within medical texts. This method efficiently converts unstructured medical data into triples while preserving the source information of each triplet, enabling convenient updates to the MedKG. (2) We constructed a medical knowledge graph (MedKG) containing partial medical information. MedKG provides healthcare professionals and medical researchers with access to comprehensive medical knowledge through inquiry, facilitating in-depth exploration and analysis. (3) We implemented a well-defined labeling system in MedKG to enhance scalability and facilitate potential integration with other disciplines' structured databases or knowledge graphs.

II. Methods

A. Experimental principle

The comprehensive workflow of our medical research is shown in Figure 1. We utilize Named Entity Recognition (NER) and Relation Extraction (RE) tasks to derive structured information about disease information, treatments, and drug composition. Subsequently, entity resolution (ER) and standardization processes are applied to integrate information from these domains into a cohesive medical knowledge graph.

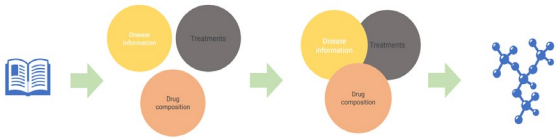

Figure 1. Workflow chart

The workflow outlined in Figure 2 begins with manual annotation and normalization of an initial training dataset, followed by fine-tuning of a large language model (LLM) specifically tailored for NER and RE tasks related to pharmaceutical entities. The resulting inference dataset is partitioned into ten segments, an essential phase for the following iterative procedure. Next, ER tasks are performed using natural language processing (NLP) techniques such as ChemDataExtractor, mat2vec, and in-house developed expert dictionaries. Following ER, superior outcomes are thoroughly curated to expand the training dataset, thus improving the model's effectiveness in future versions.

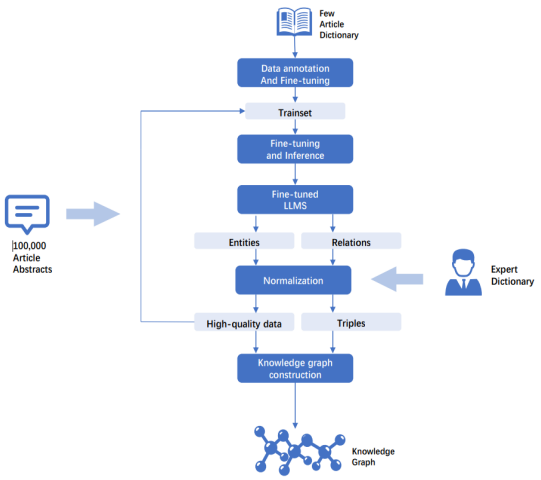

Figure 2. Pipeline of the fine-tuned LLM for knowledge graph tasks

Ultimately, the medical knowledge graph is formed using the extracted trios from the normalized results obtained after the final iteration. This systematic approach ensures the integration of disparate medical information into a structured and accessible knowledge graph, thereby facilitating comprehensive analysis and research in the medical field.

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B. Data preparation and schema design

Medical experts carefully annotated 9 different categories from 55 medical research paper abstracts as a training dataset for a large language model (LLM). These categories include basic tags such as "disease name," "treatment," or "prescription," which, although in different formats, always represent specific medical concepts in different papers. Additionally, supplemental labels include "Symptom Description," "Biological Structure/Function," "Clinical Application," "Patient Outcomes," "Treatment Methods," and "Diagnostic Methods." The "Patient Outcomes" tab summarizes quantitative results such as "Survival," "Recovery Time," and "Mortality," while the "Symptom Description" tab contains qualitative descriptions such as "Acute," "Severe," and "Severe." Taking into account the practical aspects of medical applications, we have also annotated under the "Patient Outcomes" tab essential parameters that are directly relevant to a specific "Clinical Application". For example, attributes such as the "efficacy rate" of vaccines, the "operative success rate" of surgeries, and the "duration of pain relief" of drugs are systematically annotated. And the central node "Medical Entity" is crucial, connected to nodes describing its nomenclature, composition and various properties. Specifically, "medical entity" includes "medical name", "disease name" and "abbreviation". "Biological structure/function" describes its anatomical or physiological aspects; "clinical application" refers to its actual medical use; and "patient outcomes" outlines its impact on patient health. The "Medical Entity" node also links to the "Symptom Description" node which provides additional qualitative information. In addition, the "Clinical Application" node branches into "Patient Outcomes," "Symptom Descriptions," and "Medical Domains," Suggesting additional specifications and contextual relevance for medical applications. To identify the origins of nodes and relationships, each node is associated with a Digital Object Identifier (DOI) node. By examining the shared "DOI" node neighbors at both ends of a relationship, we can pinpoint the source article from which the relationship originated.

Considering the complexity of medical sentences and the variety of terms used in abstracts, a normalization process was applied following the initial extraction. This standardization is intended to provide a consistent representation of entities with similar meanings in medical texts. For instance, terms like "cardiovascular disease" and "heart disease" were standardized to "cardiovascular disease," while terms such as "surgical intervention approach," "postoperative care approach," and "therapy-based approach" were simplified to "treatment procedure." ". Undergoing this normalization process, all annotated entities achieve consistency, which in turn facilitates the efficient training of language models (LLMs). We have collected 100,000 abstracts of medically relevant peer-reviewed research articles from authoritative medical databases, each abstract is organized in JSON format with a "DOI - text" structure to facilitate smooth processing and thorough analysis.

C. LLMs training, evaluation and inference

The training dataset includes aggregated information used for fine-tuning models such as BioMedLM, LLama2 7B, and ChatGLM 6B. After achieving high-quality outcomes from normalized inference, we repeatedly retrained the fine-tuned LLM to predict following data sets. The models were trained for 8 iterations with a single-sample batch size. Furthermore, we labeled 50 abstracts to evaluate the LLM's performance. Our assessment focuses on the model's Named Entity Recognition (NER) capabilities and explores its ability to identify potential internal relationships among sets of related elements in the Relationship Extraction (RE) task. Furthermore, during the data compilation phase, we standardized entities and evaluated the effectiveness of the LLMs in this standardization process. For tasks such as Named Entity Recognition (NER), Relation Extraction (RE), and Entity Resolution (ER), our performance evaluation utilized a comprehensive framework, focusing on metrics such as accuracy, sensitivity, and harmonic mean, while considering errors like incorrect positives and missed detections.

$$Accuracy = \frac{True\ Positives\ (TP)}{True\ Positives\ (TP)\ +\ True\ Positives\ (FP)} \quad (1)$$

$$Recall = \frac{True\ Positives\ (TP)}{True\ Positives\ (TP)\ +\ False\ Negatives\ (FN)} \quad (2)$$

$$F1 = 2 \times \frac{Accuracy \times Recall}{Accuracy + Recall} \quad (3)$$

In the domain of medical text analysis, a true positive (TP) signifies the accurate identification of an entity, while in Relationship Extraction (RE), it indicates the correct identification of a relationship between medical entities. In Entity Resolution (ER), An accurate positive denotes the exact alignment of an entity with the given schema. On the other hand, a false positive (FP) happens when the model incorrectly detects an entity, relationship, or standardization, while an overlooked positive (FN) signifies the model's failure to recognize a correct entity, relation, or schema element. Based on these definitions, we can assess each task using Equations (1), (2), and (3). After performing the evaluation, we select a fine-tuned Large Language Model (LLM) that shows optimal performance in NER and RE tasks to process 100,000 abstracts iteratively. The results of these inferences are formatted as "DOI - text - response." Thus, the fine-tuned LLM performs both entity recognition and categorization, achieving NER and RE tasks simultaneously. Additionally, all identified entities and relationships in the output are trackable, improving data integrity and usability.

D. Data matching

The accuracy and reliability of a Medical Knowledge Graph (KG) are paramount, necessitating a meticulous review and correction of inference results before constructing the graph. To standardize core medical labels, we implement a two-step process: ER-NF/A to ER-N/F. During the ER-NF/A phase, we utilize specialized medical extraction tools to extract medical terms and their corresponding acronyms from abstracts. Subsequently, we employ the MedNLP model for utilizing word embeddings to compare entities identified by both Language Models (LLMs) and extraction tools to correct and ensure the precision and accuracy of "Name" and "Acronym" correspondence. However, the classification of medical terms and their formulas is often inaccurate and challenging to discern using standard extraction tools. Hence, we perform ER-N/F by utilizing 1500 accurately classified labels as a training set to fine-tune the LLM for binary

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classification. Furthermore, we choose an extra 150 labels to assess how well the dichotomous classification performs.

For normalizing objects and connections in other labels (ER-OL), we employ clustering of word embeddings to create a specialized medical lexicon. This involves converting medical terms into vector form using MedNLP and clustering semantically similar words based on vector similarity. Specifically, we create a method for dynamic vector clustering that is based on density and dynamically forms clusters without predefined cluster numbers. This adaptive clustering method employs a threshold of similarity to establish group memberships, allowing medical entities to join or form clusters according to their closeness in feature spaces, fundamentally adapting to the inherent structure of medical data.

In the domain of medical literature analysis, we employed domain experts to categorize clusters, yielding approximately 500 key terms. These terms encompass diverse aspects such as anatomical structures, physiological phases, clinical applications, diagnostic methods, therapeutic modalities, patient outcomes, and disease descriptions. Subsequently, we applied analytical methodologies to integrate this expert-derived vocabulary with entities identified through advanced natural language processing techniques using MedNLP. Following a thorough similarity comparison process, we completed the standardization to ensure the accuracy of both entities and their relationships. It is noteworthy that the scope of "Characteristics" and "Attributes" labels often extends beyond specific medical domains and may include a wide range of related information. Consequently, we implemented a relatively flexible standardization approach for these labels, focusing on aligning content while retaining unmatched entities and relations.

To enhance the quality of our training dataset, we meticulously selected a subset of high-quality data from the normalized inference results of each iterative process, integrating this data into our training set. This high-quality Data is characterized by its precision and recall measurements, ensuring the robustness of our training regimen. Furthermore, we conducted a rigorous evaluation of the fine-tuned natural language model, trained on progressively refined datasets, to ensure optimal performance in analyzing and interpreting medical literature.

E. Knowledge graph construction

To organize the deduced outcomes into triples in the medical domain, we utilize three labels that denote the medical concept as the primary tag, where it serves as the main component, the names of other labels constitute the relationships, and their values represent the associated details. Within the structure of primary labels, "Symptom" is given the highest priority, followed by "Disease", and then "Treatment". Additionally, each main concept and its associated details are linked to the Digital Object Identifier (DOI) of the source article from which the information was extracted. This linkage enables us to trace the origin of the relationship between any two concepts by examining the overlap in their associated entities. Subsequently, we convert the triples into a Medical Knowledge Graph (MedKG) and store it using the Neo4j graph database, which also supports subgraph matching functionality. This feature aligns well with the requirement to search for specific medical concepts based on user-input criteria. To enhance accessibility to the comprehensive data, we also offer the data set available in CSV format for convenient information management.

III. Experimental results

In Table I, we present a performance comparison of Language Model Models (LLMs) including BioMedLM, ChatGLM 6B, and LLaMA2 in Named Entity Recognition (NER), Relationship Extraction (RE) and Entity Matching (ER) tasks within the medical domain. BioMedLM demonstrates significantly higher F1 scores in both Named Entity Recognition (NER) and RE tasks compared to the ChatGLM 6B and LLaMA2 7b models, indicating its superior performance in text analysis tasks related to medicine. However, there is no notable variation in their efficacy for ER activities among these models, we attribute this observation to the limited contextual memory capabilities of the LLMs. To address this challenge, we applied entity resolution to the inference outputs, resulting in substantial improvements in the Normalization (ER) activities. This enhancement method has also enhanced the performance in NER and Relation RE activities. The enhancements are due to the ER task's capability to correct erroneous inferences caused by hallucinations and to further refine the derived connections.

TABLE I. Result of NER, RE and ER through Fine-tuned LLMs.

Model	Evaluation indicators			
	Task	***Precision***	***Recall***	***F1 score***
ChatGLM 6B	NER	0.6480	0.6611	0.6506
	RE	0.5121	0.5301	0.5162
	ER	0.3766	0.3865	0.3803
LLaMA2 7B	NER	0.7280	0.7535	0.7402
	RE	0.6121	0.6352	0.6281
	ER	0.4566	0.4652	0.4601
BioMedLM	NER	0.8332	0.8211	0.8269
	RE	0.7101	0.7001	0.7086
	ER	0.4785	0.4698	0.4736
BioMedLM (Normalized)	NER	0.9322	0.9289	0.9301
	RE	0.9022	0.8892	08965
	ER	0.9100	0.8901	0.9072

In order to comprehensively evaluate the enhancements achieved at each stage of the normalization procedure, we performed ablation studies. The findings of these experiments are presented in Table II, distinctly demonstrating the contribution of each method to the enhancement of Entity Resolution (ER) tasks. The findings indicate that optimizing through the expert dictionary is the most effective method, highlighting not only its accuracy but also its comprehensive coverage of all labels except for those representing specific medical entities. The improvements observed in ER-NF/A are particularly noteworthy, as this process effectively eliminates most erroneous medical entities and exhibits a stronger impact on Named Entity Recognition (NER) than ER-OL.

TABLE II. Result of the ablation experiment in normalization.

Method	Evaluation indicators					
	NER F1	∇	***NER F1***	∇	***ER F1***	∇
BioMedLM (Normalized)	93.01	-	89.65	-	90.72	-
ER-N/F	93.01	-	83.97	-5.68	90.72	-
ER-N/F	83.65	-9.36	83.49	-6.16	85.36	-5.36
ER-OL	88.29	-4.72	80.03	-9.62	55.10	-35.62

Additionally, in Entity Resolution-Named/Formula (ER-N/F), there is a focus on correcting relationship errors between

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"Name" and "Formula," which is deemed an essential aspect of Entity Resolution (ER). Furthermore, we conducted an evaluation of the Large Language Model (LLM) classifier by utilizing a dataset of 300 manually labeled entries. The data analysis outcomes shown in Table III highlight the strong performance of the LLM in categorization tasks.

TABLE III. BINARY CLASSIFICATION PERFORMANCE THROUGH FINE-TUNED BIOMEDLM.

Model	Evaluation indicators		
	Accuracy	*F1 Score*	*Specificity*
BioMedLM	0.9586	0.9456	0.9640

The data normalization process converts the triples into a medical knowledge graph (MedKG), including 142,605 nodes and 631,772 edges, as shown in Figure 3. The graph includes 6 node types and 9 relationship types. MedKG contains a large amount of medical information, covering areas such as diseases, symptoms, drugs, patient information, etc. T This validates that our knowledge graph efficiently retains factual information and serves as a robust platform for medical researchers.

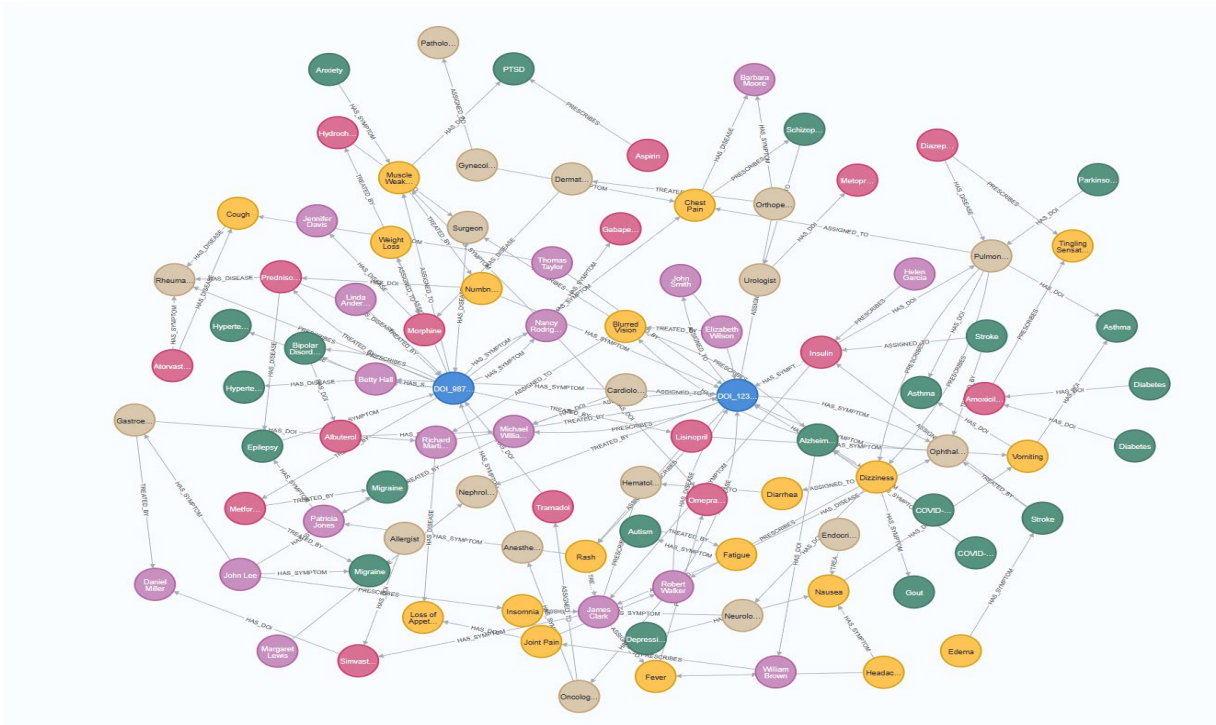

Figure 3. Local schematic diagram of MedKG

IV. CONCLUSION

In this research, we introduce a novel natural language processing (NLP) pipeline tailored for constructing knowledge graphs (KGs) aimed at efficiently extracting triplets extracted from unstructured scientific text. The unique aspect of this method lies in its capacity to fine-tune a large language model (LLM) with minimal data annotation. This enables the fine-tuned LLM to extract structured information from vast amounts of unstructured text. By bypassing predictive algorithms, this approach ensures the highest level of authenticity and traceability for the structured information. Utilizing this technique, we created an operational medicine knowledge graph (MedKG) containing medical material and related knowledge sourced from abstracts of 100,000 peer-reviewed papers. Subsequent analysis validated the effectiveness and credibility of MedKG.

Our methodology and the resulting knowledge graph possess considerable potential across multiple dimensions. Firstly, by deepening the extraction of structured data, including entire research articles, we aim to create a more thorough and comprehensive knowledge graph. This entails not only broadening the range of data analysis but also enhancing methods to encapsulate subtleties in intricate academic literature. Furthermore, improving identifiers for entities in our framework facilitates finer data classification, including the addition of granular attributes like preparation parameters or precise parameters. This enhancement substantially increases the level of detail and usefulness of the knowledge graph. In addition, the flexibility of our NLP process demonstrates its adaptability across various scientific fields, providing a foundation for building specialized knowledge networks. Finally, integrating MedKG integrating with existing knowledge graphs enables the creation of interconnected and holistic datasets, fostering cutting-edge research and application development in medical science and other fields.

ACKNOWLEDGMENT

This work was supported by the Natural Science Foundation of Zhejiang Province (LTGS23F030003).

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